

# TACT: Trust-Anchored Confidence Tempering for Self-Consistency Voting

*Signed, analytically-derived, optionally label-free trust in a frozen LLM's confidence channel — with exact self-consistency anchors and a pre-registered falsification protocol.*

|                                                                                                                                                  |                                                                                                                     |                                                                                                                  |                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>1.000 vs 0.807 floor</b><br>label-free accuracy on an anti-correlated channel ( $\kappa = -0.6$ ); every published protocol sits at the floor | <b>1.000 vs 0.965 oracle</b><br>rank invariance beats the entire raw-value weight family under monotone compression | <b>0.940 +79 / -0</b><br>label-free per-group voting vs the 0.808 floor; zero losses to SC over 600 paired items | <b>4 / 4</b><br>pre-registered falsifiers survived, incl. the published CISC-devT protocol and a dev-picked signed grid |
|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|

## PROBLEM

Confidence-weighted self-consistency (CISC and its successors) extracts large gains from a frozen LLM's self-reported confidence — until the channel is miscalibrated in *direction*. Every published weighting scheme is structurally monotone **increasing** in confidence, so an **anti-correlated** channel poisons the vote rather than informing it. Binary dev-set gates (e.g. on ECE) survive by discarding the channel entirely, wasting genuinely discriminative signal. No published method can *estimate the sign* of the confidence–correctness association, and none can do so without labels.

## CONTRIBUTIONS

### C1 Signed, analytically-tempered confidence weighting

We replace CISC's fixed exponent with a derived one:  $\gamma = \text{clip}(L(\text{shrink}(D, SE)))$ , where  $D$  is the pooled van Elteren Somers'  $D = 2 \cdot \text{WQD} - 1$  of the channel — a pure within-item rank statistic, invariant to every strictly monotone distortion of the confidence scale — shrink is positive-part James–Stein with a significance floor, and  $L$  is the Bayes-discriminant link with a mixture-variance correction. The map carries exact anchors: inside the dead zone the vote is **bit-identical to plain self-consistency** (a shared code path — no paired test can distinguish them), and with the log-value feature map the family reproduces CISC-power exactly. Voting with  $w_i = \exp(\gamma \cdot \phi_i)$  on standardized within-item rank scores makes the whole method rank-invariant: under monotone compression it beats the *oracle over the entire raw-value weight family* (1.000 vs 0.965; Fig. 2).

### C2 Label-free estimation of the signed channel reliability

The crowdsourcing lineage (Dawid–Skene; spectral meta-learners) estimates reliability from cross-annotator covariance; a single exchangeable confidence channel offers none. We estimate signed discrimination from *agreement pseudo-labels* (dedup-weighted plurality per item) and prove a class-conditional-noise attenuation identity  $E[D_g] = (1 - 2\rho) \cdot D_{\text{true}}$ : the estimate can only *under-trust*, never mis-sign, whenever the pair-weighted plurality-error rate stays below  $\frac{1}{2}$ . A split-half agreement inversion de-attenuates conservatively, and sign-aware alarms return the method to plain SC when identifiability is threatened. On the coupling sweep the label-free variant matches the 200-label variant nearly point-for-point — including full recovery of **negative** channels ( $\kappa = -0.6$ : **1.000 label-free** vs the 0.807 floor of every published protocol; Fig. 1). The honest boundary is stated: under a confident verbatim echo the sign is information-theoretically ambiguous; the alarms detect the verbatim case and refuse, and  $\sim 50$  labels (sign only) restore full operation.

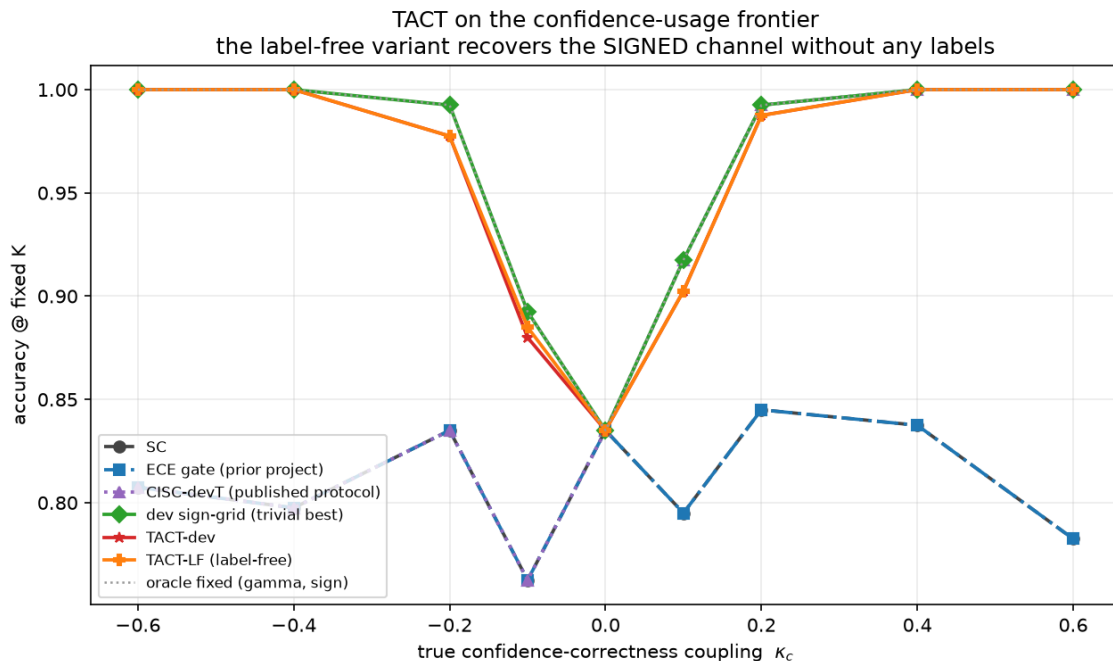
### C3 An impossibility result for per-item adaptation — and the covariate-structured escape

When the per-item coupling  $\kappa_q$  is i.i.d. with no observable covariate, per-item label-free adaptation is closed: (i) a *self-reinforcement identity* — any monotone map from an item's own agreement statistic to an exponent reweights toward the plurality on both branches, collapsing to SC (97.5% agreement; residual flips net-harmful); (ii) a *winner's curse* — on exactly the plurality-wrong items where a flip could win, the observable sign opposes the truth 96% of the time; (iii) *two-world unidentifiability* —  $\{\kappa > 0, \text{minority correct}\}$  and  $\{\kappa < 0, \text{plurality correct}\}$  induce identical observable laws. The per-item oracle (0.983) is therefore unreachable, and TACT's dead zone degrades to *exactly* SC there (zero discordant pairs; Fig. 3, right). When heterogeneity is instead indexed by an observable covariate — domain-dependent calibration, the realistic case — the same estimator run per group recovers each group's signed coupling (label-free:  $\{+2.0, 0.0, -2.0\}$ ) and cracks the floor that provably binds every global policy: **0.940 label-free vs the 0.808 floor**, within 0.007 of the per-item oracle, with zero losses to SC ( $+79/-0$ ,  $p = 3.3 \times 10^{-24}$ ; Fig. 3, left).

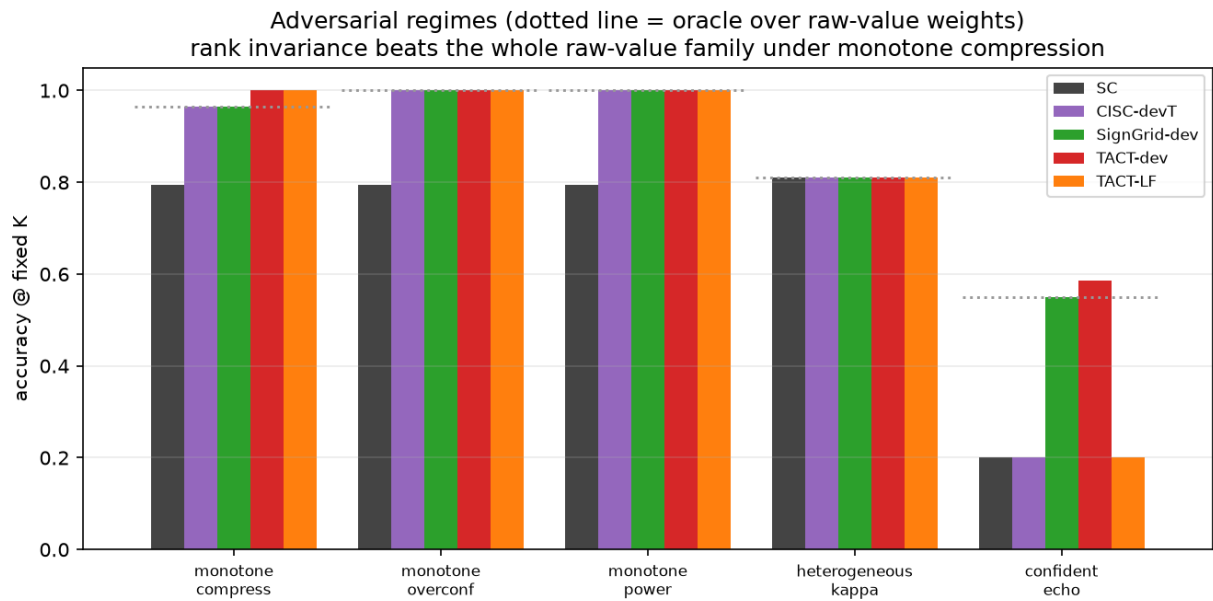
### C4 A pre-registered falsification protocol with the strongest baselines included

Four falsifiers were fixed before implementation — including the two designed to kill the method: the *published* dev-calibrated protocol (CISC-devT, whose tuned temperature already interpolates  $SC \leftrightarrow CISC$ ) and a trivial dev-picked signed exponent grid. All four survived, and the honest margins are reported: against the signed grid the net advantage concentrates in exactly three cells — monotone distortion (+0.035), confident echo (+0.035), and label-free operation, which no grid can perform. As a matched pair of honest outcomes, the same protocol applied to our preceding system (RLEV-VoI, redundancy-discounted voting) *fired* four of five falsifiers; that system is reported as a negative result whose post-mortem motivated this work.

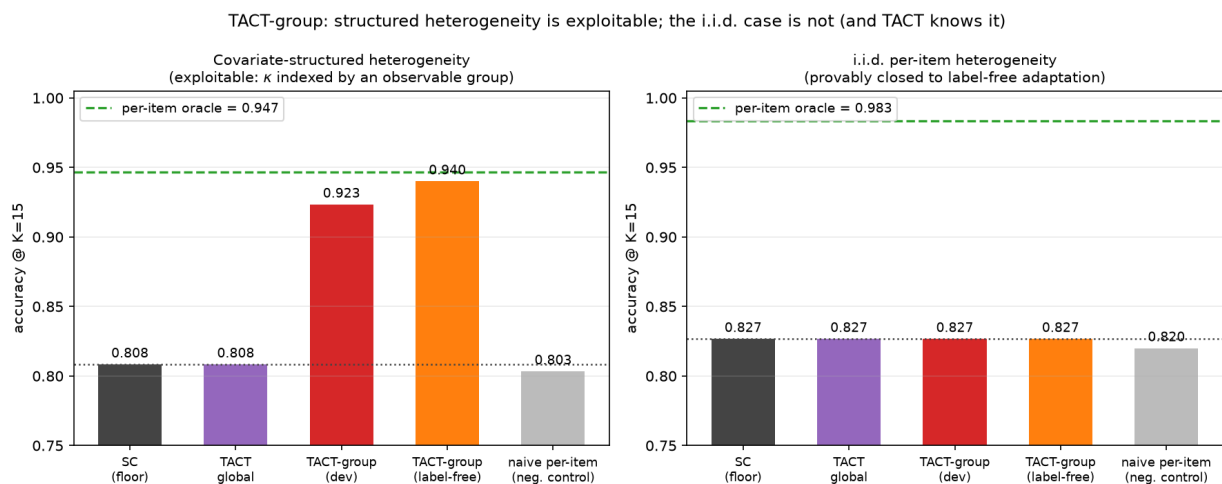
## FIGURES



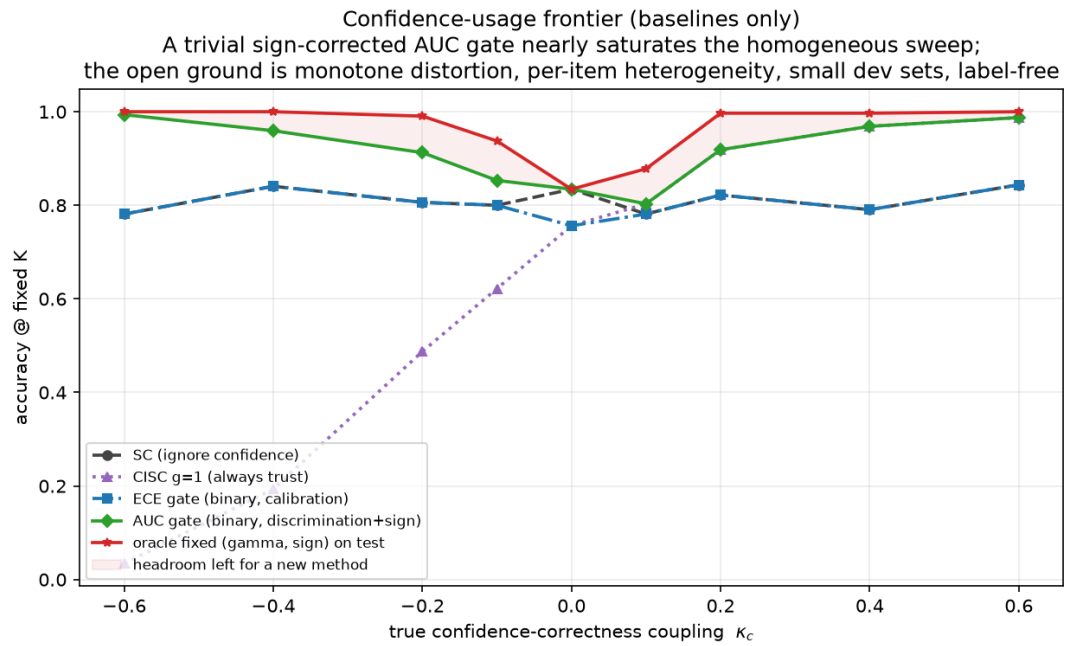
**Figure 1 — The confidence-usage frontier.** Published protocols (CISC-devT, the ECE gate) sit at the self-consistency floor across the entire negative half-axis; TACT-dev and the fully label-free TACT-LF track the oracle over the whole sweep, including signed recovery at  $\kappa < 0$ . 400 paired items per cell,  $K = 15$ .



**Figure 2 — Adversarial regimes.** Dotted line: the oracle over raw-value weights. Rank invariance beats that entire family under monotone compression (1.000 vs 0.965). In the confident-echo cell the labeled variant counters the poison with a negative exponent ( $\gamma = -1.20$ ; 0.585 vs SC 0.200) while the label-free variant alarms and refuses — its conditional guarantee working as stated.



**Figure 3 — Structured vs i.i.d. heterogeneity.** Left: with an observable covariate, per-group TACT (label-free, 0.940) approaches the per-item oracle (0.947) from a floor of 0.808 with zero losses to SC. Right: the provably-closed i.i.d. cell — every legitimate method sits at the floor and TACT degrades to exactly SC; the naive self-referential method (negative control) lands slightly below it.



**Figure 4 — The pre-measured problem statement.** Baseline headroom mapped *before* TACT existed: a trivial sign-corrected AUC gate nearly saturates the homogeneous sweep, isolating the only cells where a new method could win — monotone distortion, heterogeneity, small dev sets, and label-free operation. TACT's claims live exactly there.

## FALSIFIERS

| ID | Falsifier (pre-registered)                                            | Outcome                                                       |
|----|-----------------------------------------------------------------------|---------------------------------------------------------------|
| F1 | TACT-dev below the best fixed- $\gamma$ CISC at $\kappa = +0.6$       | survived — 1.000 vs 1.000                                     |
| F2 | Either variant significantly below SC anywhere on the sweep           | survived — bit-identical to SC at $\kappa = 0$                |
| F3 | Label-free variant fails to beat the binary ECE gate on sweep average | survived — 0.954 vs 0.811                                     |
| F4 | CISC-devT or the dev signed grid matches TACT everywhere              | survived — distortion and echo cells are unreachable by grids |

**Evidence status.** All quantitative claims are on a synthetic-oracle harness (paired trace pools, McNemar tests, 400–600 items per cell) whose adversarial regimes lie outside the estimator's working model; mechanism-recovery and accuracy claims are reported separately to limit circularity. Validation on real LLM traces is the remaining step; the cached-trace runner is committed (experiments/run\_real\_api.py).

Repository: vito1317/adaptive-reasoning-consensus (private) · spec docs/SPEC-TACT.md · full report docs/REPORT-TACT.md · 76 tests. Every number traceable to results/\*.json; figures reproducible from the committed scripts.